

High Integrity GNSS Navigation and Safe Separation Distance to Support Local-Area UAV Networks

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ABSTRACT

This paper develops an methodology to estimate safe separation distance for UAVs which share the same source of guidance and LADGNSS differential corrections.

1.0. INTRODUCTION

Recent technological improvements and increased military utilization have proven the operational viability of unmanned aerial vehicles (UAVs) and made them attractive for a wide range of potential civil applications. In particular, the capability and low cost of small UAVs have led to a dramatic growth in potential commercial applications over the past few years as well as a major interest in the media. While most current uses of UAVs are remotely piloted by humans, future UAVs will operate autonomously and coordinate their activities in large groups. Autonomous UAVs networks guided by central controllers will safely and routinely perform various missions such as surveillance, reconnaissance, and data collection [1].

Prior work [1, 2] has proposed a concept of UAV network operations that use differential corrections from Local-Area Differential GNSS (LADGNSS) to provide increased accuracy, safety, and reliable navigation to UAVs within 5 to 100 km of the controller station. LADGNSS supports low cost for commercial applications and high integrity to allow UAVs to operate in close proximity to each other while minimizing collision risk. Local-area UAV networks can perform their tasks safely as long as UAV locations within the network are

monitored by the controller to insure that safe separation distances are maintained from other vehicles and ground obstructions. The controller, including a LADGNSS reference station, receives GNSS signals and positions of UAVs, estimates errors and separation distances, and broadcasts differential corrections and optimized guidance for precise and safe operation of UAVs.

This paper presents a methodology to estimate the desired minimum separation distance that should be defined to prevent collisions by being utilized for guidance of UAVs. Separation distance can be derived by modeling mainly two sources of position uncertainty that could potentially lead to collisions: (i) navigation error, or the residual error in the estimated position (compared with the true position), which cannot be completely eliminated by LADGNSS; and (ii) flight technical error, or error in following the path defined by the guidance source, which depends on the controller's performance and external disturbances such as wind.

This paper estimates these errors using standard error models of LADGNSS developed for manned aircraft and bounds derived from empirical data collected from UAV flight tests.

2.0. LOCAL AREA UAV NETWORK OVERVIEW

Figure 1 illustrates the concept of local-area UAV network operations developed in [1,2]. The control station shown at the lower left is the source of local-area differential GNSS corrections and integrity information as well as real-time guidance for each UAV in the network. The LADGNSS and guidance information are separate data messages combined in the same outbound

transmission. The guidance function also requires feedback from each UAV in real time, including its current position and velocity. Therefore, a two-way datalink is required and is used to relay GNSS information (such as position-domain protection levels) as well as position and velocity from UAVs to the control station.

Safe separation distance

3.0. MODELING(?) METHODOLOGY

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The

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3.1. RESIDUAL TROPOSPHERIC ERROR MODLEING

As each UAV is at a different altitude than the ground reference stations, a differential tropospheric delay correction is required. For GBAS applications, the vertical tropospheric correction in meters for a user at altitude Δh meters above the reference station can be given by

$$TC_{vert} = 10^{-6} N_R h_0 (1 - e^{-\Delta h / h_0})$$

with

N_R = refractivity index transmitted by ground subsystem
[].

h_0 = tropospheric scale height transmitted by ground subsystem (in meters) [].

For a satellite at elevation angle θ , the slant (i.e., along the actual path between satellite and receiver) tropospheric correction is defined as follows:

$$TC = \frac{N_R h_0 \times 10^{-6}}{\sqrt{0.002 + \sin^2(\theta)}} (1 - e^{-\Delta h / h_0})$$

Using the refractivity uncertainty transmitted by ground subsystem (σ_N), the residual tropospheric error bound for LAAS is also defined by:

$$\sigma_{tropo} = \frac{\sigma_N h_0 \times 10^{-6}}{\sqrt{0.002 + \sin^2(\theta)}} (1 - e^{-\Delta h / h_0})$$

For the LADGNSS errors among in-network UAVs, we are interested in the differential tropospheric delay as observed at an $UAV_a \Delta h_a$ meters above the reference

station and at another $UAV_b \Delta h_b$ meters above the reference station. As the correlated tropospheric error between two UAVs is canceled out, the residual tropospheric error bound between UAVs is computed as:

$$\begin{aligned} \sigma_{tropo, between UAVs} &= \sigma_{tropo}(UAV_b) - \sigma_{tropo}(UAV_a) \\ &= \frac{\sigma_N h_0 \times 10^{-6}}{\sqrt{0.002 + \sin^2(\theta)}} (e^{-\Delta h_b / h_0} - e^{-\Delta h_a / h_0}) \end{aligned}$$

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3.2. RESIDUAL IONOSPHERIC ERROR MODLEING

Ionospheric temporal and spatial decorrelation can lead to residual range errors to a LADGNSS user. The temporal decorrelation refers to the ionosphere pseudorange delay changes over time at the same point in the ionosphere. If users use same carrier-smoothing time constants with reference station, the temporal decorrelation does not lead to steady state residual range error. As we assume that carrier-smoothing time constants are matched between reference station and in-network UAVs, the spatial decorrelation, where the delay changes from point to point in the ionosphere at a given time, is only interested. The residual ionospheric spatial decorrelation error model for UAVs is an extension of the original work, which derived a simplified model (σ_{iono}) of residual user range error due to spatial decorrelation for GBAS. The original model was for ionospheric decorrelation errors between a fixed reference location and approaching aircraft. Therefore, in this paper, a generalized version of the ionospheric model which considers both UAVs as moving is developed. The differential carrier-smoothed ionosphere delay between UAV_a and UAV_b at epoch k ($\Delta \hat{I}_k$) can be written as:

$$\Delta \hat{I}_k = \frac{\partial I_r}{\partial x_r} x_r - 2(\tau - T_s) \frac{\partial I_b}{\partial x_b} v_{b,k-1} + 2(\tau - T_s) \frac{\partial I_a}{\partial x_a} v_{a,k-1}$$

where x_r is range distance (in meters) between UAV_a and UAV_b ; $\partial I_r / \partial x_r$ is the spatial ionospheric gradient between UAVs; τ is the carrier smoothing time constant (in seconds); T_s is the sampling time (in seconds); $\partial I_a / \partial x_a$ and $\partial I_b / \partial x_b$ are the spatial ionospheric gradients in the direction of motion of the UAV_a and UAV_b respectively; and v_a and v_b are the velocity of the UAV_a and UAV_b respectively. This differential delay consists of the spatial ionospheric decorrelation between the UAVs, which express as $\partial I_r / \partial x_r \cdot x_r$, and the smoothing effects. As smoothing effects are functions of the velocity of the UAVs and the spatial ionospheric gradients in the direction of motion of the UAVs, ionospheric model for UAVs has been more complex when UAVs are moving.

To derive the model, two conditions are assumed. First, ionosphere is a spatially linear ionospheric decorrelation with gradient of G as shown in Figure ?. In the figure, α represents an angle between ionospheric front with gradient of G and line of sight between UAVs. The θ_a and θ_b are angles between line of sight between UAVs and a vector of velocity of the UAV_a and UAV_b respectively. In this condition, $\partial I_r / \partial x_r$, $\partial I_a / \partial x_a$ and $\partial I_b / \partial x_b$ can be obtained as

$$\frac{\partial I_r}{\partial x_r} = G \cos \alpha$$

$$\frac{\partial I_a}{\partial x_a} = G \cos(\alpha + \theta_a) = G(\cos \alpha \cos \theta_a - \sin \alpha \sin \theta_a)$$

$$\frac{\partial I_b}{\partial x_b} = G \cos(\alpha + \theta_b) = G(\cos \alpha \cos \theta_b - \sin \alpha \sin \theta_b)$$

Therefore, the differential carrier-smoothed ionosphere delay can be formulated as

$$\Delta \hat{I}_k = G \left[\cos \alpha (x_r - 2\Delta x_{b,k-1} \cos \theta_b + 2\Delta x_{a,k-1} \cos \theta_a) + \sin \alpha (2\Delta x_{b,k-1} \sin \theta_b - 2\Delta x_{a,k-1} \sin \theta_a) \right]$$

Using the trigonometric function, the differential delay can be simply express as

$$\Delta \hat{I}_k = A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \sin(\alpha + \varphi),$$

$$\sin \varphi = \frac{A}{\sqrt{A^2 + B^2}}$$

To conservatively bound the errors, we assumed second condition, which the ionospheric slope is located in the direction that maximizes the differential delay between UAVs. On the other hands, we determine the $\sin(\alpha + \varphi)$ in Eq.? as one. Using one-sigma value for nominal ionosphere vertical spatial gradient (σ_{vig}), the residual ionospheric error bound between UAVs is derived as

$$\sigma_{iono, between UAVs} = F_{pp} \times \sigma_{vig} \sqrt{\left(x_r - 2(\tau - T_s) v_{b,k-1} \cos \theta_b + 2(\tau - T_s) v_{a,k-1} \cos \theta_a \right)^2 + \left(2(\tau - T_s) v_{b,k-1} \sin \theta_b - 2(\tau - T_s) v_{a,k-1} \sin \theta_a \right)^2}$$

where F_{pp} is the vertical-to-slant obliquity factor for a given satellite.

$$F_{pp} = \left[1 - \left(\frac{R_e \cos \theta}{R_e + h_I} \right)^2 \right]^{-\frac{1}{2}}$$

with

R_e = radius of the earth = 6378.1363 km.

h_I = ionospheric shell height = 350 km

θ = the elevation angle of satellite

If this model applies the ionospheric decorrelations between a fixed reference location and approaching aircraft which are considered for development of original model, which v_a is zero, v_b is v_{air} , θ_a is 0° , and θ_b is 180° , we can obtain the formula as following.

$$\sigma_{iono} = F_{pp} \times \sigma_{vig} \times (x_r + 2 \times (\tau - T_s) \times v_{air})$$

This equation is same with the ionospheric model for GBAS. Therefore, the model developed in this paper is a generalized version of residual ionospheric error model including the original .

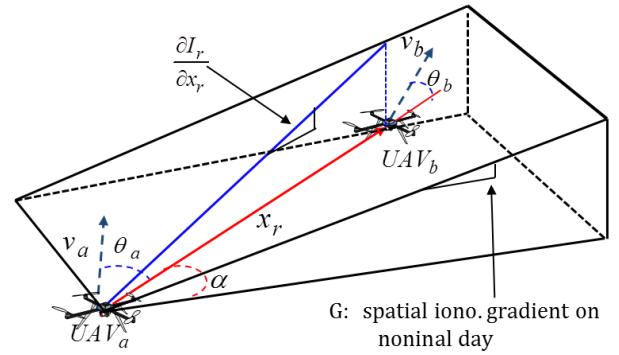


Figure 1

3.3. AIRBORNE MULTIPATH MODLEING

3.4. FLIGHT TECHNICAL ERROR MODLEING

4.0. CONFIGURATION

5.0. RESULT

5.1. AIRBORNE MULTIPATH MODLEING

5.2. FLIGHT TECHNICAL ERROR MODLEING

5.3. SEPARATION DISTANCE

ACKNOWLEDGMENTS

REFERENCES

- [1] Lee, J., Jung, S., Bang, E., Pullen, S., and Enge, P., “Long Term Monitoring of Ionospheric Anomalies to Support the Local Area Augmentation System,” *Proceedings of ION GNSS 2010*, Portland, OR, Sept. 21-24, 2010, pp. 2651-2660.